(a)

1	2	3	4

5	6

```
struct Node {
    int val;
    Node*  next;
};

Node* llrec(Node* in1, Node* in2)
{
    if(in1 == nullptr) {
        return in2;
    }
    else if(in2 == nullptr) {
        return in1;
    }
    else {
        in1->next = llrec(in2, in1->next);
        return in1;
    }
}
```

NULL 2

(1, 5) → (5, 2)

1 → ?

5 → 2

2 →

6 → 4

(2 → 6)

(6, 3)

(4, NULL)

$\boxed{1\ 5\ 2\ 6\ 4}$

part (b)

in1 = nullptr

in2 = 2

(nullptr , 2)

⇓

if statement

⇓

return in2 ⇒ $\boxed{2}$